Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning

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01 / BACKGROUND

We know the past. Can we anticipate future cases? Weekly bulletins from Mexico's National Epidemiological Surveillance System [1] compile newly notified cases reported by health services. We use this history to forecast **notifications, not clinical incidence.**

02 / OBJECTIVES

Forecast weekly notified cases of **depression (F32), Alzheimer's (G30) and Parkinson's (G20)**, conditions of neuropsychiatric interest, over a **52-week horizon.** Measure error to explore their potential use in planning services, staffing and budgets.

ICD-10: International Classification of Diseases, tenth revision.

03 / METHODS

Retrospective study, 2014-2026. Each time series follows weekly notified cases by condition, area and sex, allowing forecasts to reflect local patterns, not just national totals.

333 series = 3 conditions × 37 areas × 3 strata
Areas: 32 states, the country and 4 state groups.
Strata: men, women and both sexes combined.

Results: both sexes combined; three conditions nationally and depression in 32 states. National forecasts are not the sum of state forecasts.

How do we forecast?

We compared four statistical and **machine learning (ML)** methods to capture different patterns. **Prophet** [2] models trends and cycles; **DeepAR** [3] learns patterns through a neural network. **Ensemble** combines forecasts; **Stacking** learns how to combine them. We use a selected model for each series: nationally, DeepAR for depression and Prophet for Parkinson's and Alzheimer's.

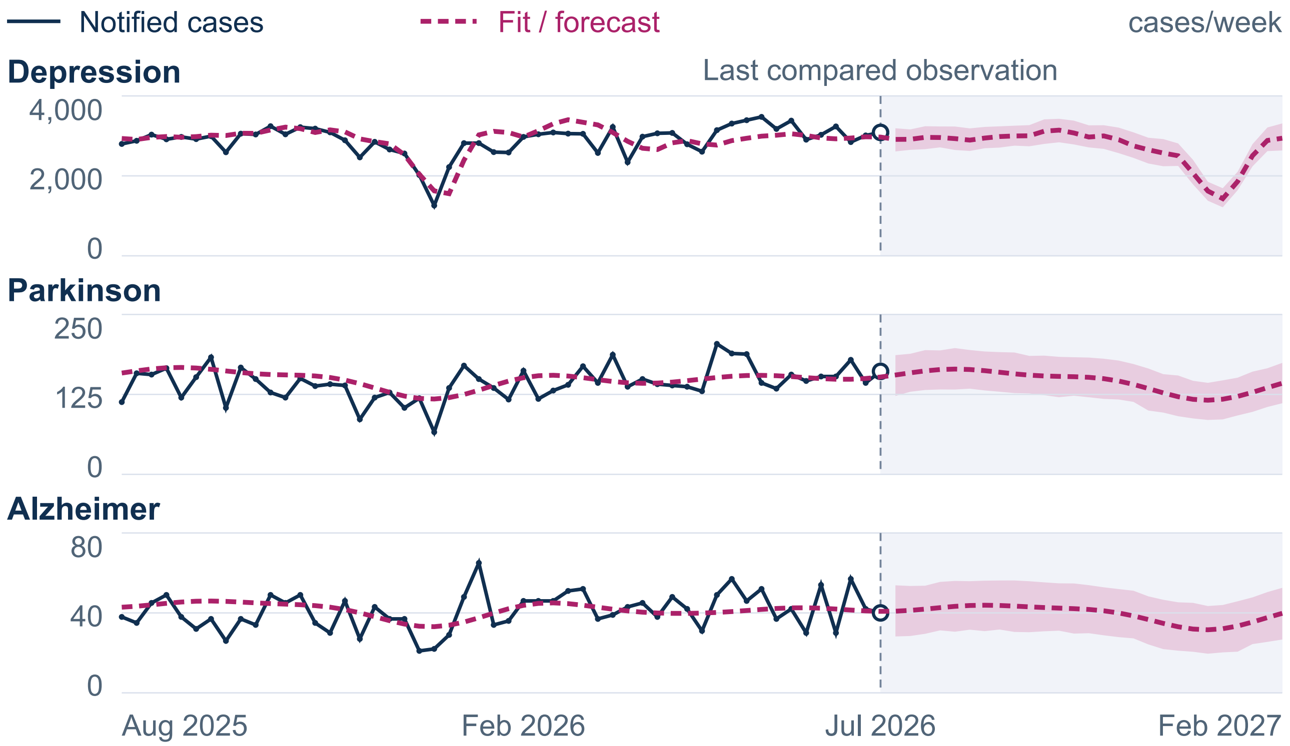
04 / EVALUATION & RESULTS

We compared forecasts with 2026 notifications: **epidemiological weeks 7 to 31 (25 weeks)**. For weeks **24 to 31**, we evaluated predictions saved before those weeks.
We used two metrics to evaluate forecast performance:
sMAPE (symmetric mean absolute percentage error): compares relative error across series; lower is better, 0% is an exact match.
MAE (mean absolute error): average difference in cases per week, whether forecasts are too high or too low.

Table 1. National errors

Condition	sMAPE 25 weeks	sMAPE 8 weeks	MAE 25 weeks	MAE 8 weeks
Depression	**8.7%**	**5.0%**	**260.2**	**153.8**
Parkinson	**10.6%**	**6.7%**	**16.5**	**10.3**
Alzheimer	**14.4%**	**17.8%**	**6.2**	**7.4**

Figure 1. History and forecast



History: retrospective fit. Model band; coverage not evaluated.

DEPRESSION · NATIONAL TOTAL

-1.6% below the reported total

74,675 Forecast | **75,859** Notified

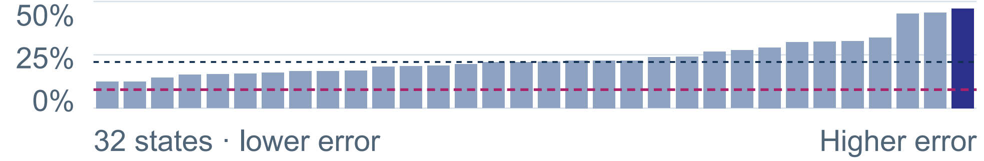
Difference in 25-week totals, not weekly error.

Depression by sex · 2026 cumulative, week 31

Notified cases; not forecasts or individual risk.

Figure 2. Depression · sMAPE, 25 weeks

National	Median	Max. Tlaxcala
8.7%	**21.6%**	**46.7%**



National accuracy does not guarantee local accuracy. With few notified cases, small errors can represent large relative differences. Check each state's error before planning its resources.

05 / CONCLUSIONS

From counting cases to preparing decisions. Forecasts could inform staffing and appointment planning by showing when and where more notifications are expected. Territorial differences help identify where to check reliability, not how much funding to allocate. Their use requires local prospective validation; usefulness for services has not yet been evaluated.

LIMITATIONS. Care demand and clinical benefit not measured; not fully prospective. Age and transfer beyond Mexico unassessed; possible underreporting.

[1] DGE. Boletín Epidemiológico, SINAVE. Secretaría de Salud, México; 2026.
[2] Taylor SJ, Letham B. Forecasting at scale. Am Stat. 2018;72:37-45.
[3] Salinas D, et al. DeepAR. Int J Forecast. 2020;36:1181-1191.

Methods & data
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